

Similar Accuracy, Opposite Failure: A Failure-Mode Decomposition of KV-Cache Eviction in Multilingual Long-Context Retrieval

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Abstract. KV-cache eviction is standardly evaluated by average benchmark accuracy, a statistic that treats a wrong answer and a withheld answer as the same event. Using the present-needle and absent-needle tasks of the multilingual ONERULER benchmark (two 7–8B models, three eviction policies, two budgets, two compression protocols, seven languages; 29,400 generations over 1,400 unique 32K-token contexts), we decompose retrieval into a *detection* decision—answer or abstain—and *identification* precision given an answer. Because the two tasks differ in composition (one needle vs. four distractors), we read hit and false-alarm rates as answer rates on the two tasks and d' , c as descriptive indices. Under the query-agnostic protocol the criterion index moves conservative in all 12 model $\times$ policy $\times$ budget conditions (paired bootstrap CIs exclude 0), and the sensitivity index falls in 11/12: degradation is structured, not graceful. The decomposition exposes a dissociation that averaged accuracy cannot see: at a 25% budget, Llama-3.1-8B under key-norm eviction and under window eviction post present-task accuracies of 0.21 and 0.25, yet the former is correct on 34% of the trials on which it answers and the latter on 99%—one policy fails by emitting wrong values, the other by falling silent. Classifying every emitted value against the item’s own input shows the wrong values are predominantly corrupted copies, not inventions: under key-norm eviction at 25%, 272 of the 282 Llama answers lacking the true value are *near-copies* of it (263 are truncations of its digits), and on absent trials 89% of the values Llama emits are exact or near-copies of a distractor. Wholesale invention is a minority mode but not a negligible one—13.5% [10.3, 17.5] of Llama’s answered absent trials contain at least one value found nowhere in the input, though only 1.4% consist of ungrounded values alone—and Qwen invents far more: novel values reach 26% of its answered present trials. A query-aware replication on all present-trial conditions shows the KNorm–StreamingLLM precision contrast persists under query visibility—though the matched-accuracy pairing does not recur—while SnapKV’s ranking is a protocol fact: its accuracy rises by up to +0.75 when the policy sees the query. A per-language audit yields two findings about the benchmark itself: (i) the Korean 32K haystacks emitted by the ONERULER generator are $4.7\times$ redundant (341 unique sentences; one repeated $171\times$), and content-based policies rank Korean best exactly where position-based eviction ranks it near-worst—consistent with redundancy exploitation, though we have not tested this causally; and (ii) the official absent-needle scorer credits any response containing the language’s “none”-word, including responses that first list the four distractor values—inflating Llama’s Korean score by 0.71 while moving Qwen’s by 0.01, i.e., the benchmark applies different effective criteria to different models. We show these defects can flip the sign of cross-language dispersion conclusions, and we recommend that compression evaluations report present/absent operating points with strict scoring rather than a single averaged accuracy.

1 Introduction

KV-cache eviction has become a default lever for serving long-context language models: retain a budgeted subset of the key–value entries produced during prefill, discard the rest, and accept what the literature typically describes as a small accuracy cost (Xiao et al., 2023; Zhang et al., 2023; Li et al., 2024; Devoto et al., 2024). The cost is almost always quantified the same way—a delta in averaged task accuracy on a retrieval or QA benchmark—and when the delta is small the method is declared safe.

Average accuracy, however, is a one-dimensional summary of a two-dimensional behavior. A retrieval query admits three outcomes, not two: the model can return the correct value, return a wrong value, or decline to return a value at all. Two compression policies can post the same

accuracy while distributing the remaining probability mass in opposite ways—one answering confidently and wrongly, the other abstaining. For a deployed system these are not interchangeable failures. A wrong value flows silently into downstream computation; an abstention is visible, triggers fallback logic, and costs only coverage. Whether compression pushes a model toward one failure or the other is precisely the information a single accuracy number destroys.

Signal detection theory (SDT) provides the standard vocabulary for separating these components (Green and Swets, 1966). If the model faces a mixture of trials on which the queried needle is *present* and trials on which it is *absent*, then the decision to produce a value at all is a detection response with a hit rate H (answering when the needle is present) and a false-alarm rate FA (answering when it is absent). Sensitivity $d' = z(H) - z(FA)$ measures how well the model discriminates presence from absence; the criterion $c = -\frac{1}{2}(z(H) + z(FA))$ measures how willing it is to

Code, per-item records, and analysis scripts: <https://github.com/elad-simbalista/kv-cache-eviction>.

answer at all. Orthogonally, *identification precision*—the probability that the value is correct given that the model chose to answer on a present trial—measures the quality of the answers that do get produced. The ONERULER benchmark (Kim et al., 2025) comes close to providing an SDT experiment: its `niah_single` task places the queried needle in the haystack, its `niah_none` task withholds it, and the instructions in both explicitly license a “none” response. To our knowledge no compression evaluation has used this structure. We use it with a caveat stated up front and carried through the analysis: the two tasks differ in more than target presence—the present task contains only the queried needle, the absent task contains four distractor needles (§3)—so we read H and FA as answer rates on the two tasks, and d' , c as descriptive indices rather than identified sensitivity and criterion.

We apply the decomposition to three eviction policies (KNorm, SnapKV, StreamingLLM) on Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct-1M across seven languages at 32K tokens, under both a query-agnostic and a query-aware compression protocol, with a response coding that is independent of the benchmark’s scorer. Four findings:

1. Compression moves the criterion index conservative in every tested condition, and the sensitivity index falls in nearly all. In all 12 model $\times$ policy $\times$ budget conditions under the query-agnostic protocol the criterion index moves toward abstention ($\Delta c > 0$, paired bootstrap CIs excluding 0), and in 11/12 the sensitivity index drops as well. “Graceful degradation” and “pure criterion shift” are both wrong as summaries; the truth is a structured movement in a two-dimensional operating space (Fig. 1).

2. Policies with similar accuracy sit at opposite ends of a precision–abstention trade-off, and the wrong values are corrupted copies, not inventions. At a 25% budget on Llama, KNorm (present-task accuracy 0.21) is correct on 34% of the trials on which it answers; StreamingLLM (0.25) on 99%. The ordering $\text{KNorm} < \text{SnapKV} < \text{StreamingLLM}$ in precision-when-answering holds in every condition on both models (Table 1). Classifying every emitted value against the input (exact copy / near-copy / entirely novel; Table 2) shows what the wrong values are: on absent trials, exact distractor copying is near-universal at baseline and under position-based eviction, and under KNorm at 25% half of Llama’s answers become *near-copies* of a distractor; 13.5% [10.3, 17.5] contain at least one value found nowhere in the input, but only 1.4% consist of ungrounded values alone. On present trials, 272 of the 282 Llama KNorm-25% answers that lack the true value are near-copies of it, 263 of them truncations of its digits. Key-norm eviction corrupts values in place; invention is the minority mode on both models and rare on Llama.

3. Policy rankings and failure composition depend on query visibility; SnapKV’s position is a protocol fact. Rerunning every present-trial condition under a query-aware variant, SnapKV becomes near-lossless (accuracy 0.90–0.92

at both budgets on both models; +0.75 [0.72, 0.77] on Qwen at 25%), while the KNorm-vs-StreamingLLM precision contrast persists (0.23/0.06 vs. 0.90/0.62 at 25%), though the matched-accuracy pairing itself does not recur under this protocol. Comparisons of eviction policies are not portable across protocols.

4. The benchmark itself contributes two artifacts, and they are large enough to flip conclusions. The Korean 32K haystacks produced by the ONERULER generator at its default settings are degenerate—341 unique sentences, one repeated 171 times, $4.7\times$ mean redundancy—and content-based eviction appears to benefit: Korean is Llama’s *worst* language uncompressed and becomes its *best* under KNorm and SnapKV, while remaining near the bottom under content-blind StreamingLLM (an observational pattern consistent with redundancy exploitation; we have not tested it causally). Separately, the official `niah_none` scorer credits any response containing the language’s none-word as a substring, so a response that lists the four distractor values and appends the none-word scores as a correct rejection. On Llama this inflates Korean by 0.71 and Swahili by 0.15; on Qwen the largest gap is 0.09. The same scorer thus applies materially different effective criteria to different models. We show by leave-one-language-out analysis that a single defective split is sufficient to invert whether compression appears to amplify or suppress cross-language disparity.

2 Related Work

Eviction policies. StreamingLLM retains attention sinks plus a recency window (Xiao et al., 2023); H2O accumulates attention mass (Zhang et al., 2023); SnapKV scores prefix tokens by attention from a trailing observation window (Li et al., 2024); KNorm exploits the negative correlation between key ℓ_2 norm and received attention (Devoto et al., 2024). We use the `kvpress` implementations (Jégou et al., 2024).

Costs of compression. A growing literature disputes the “negligible loss” framing. Yuan et al. (2024) benchmark what compression gives up across long-context tasks; Liu et al. (2025) test retention of fundamental abilities; Yang et al. (2024) document safety breaches, incoherence, and hallucination caused by eviction and mitigate them with mixed-precision retention; A. Chen et al. (2026) show that eviction is biased across instructions, leaking system prompts, and that the effects depend heavily on policy and model; Anantharayanan et al. (2026) argue from attention dynamics that retaining a KV pair does not guarantee semantic accessibility, and document a hallucination cliff near 90% compression that tracks cross-head erasure of answer-critical tokens. Our contribution is complementary to all of these: not a new task on which compression fails, but a decomposition showing that the *shape* of failure—abstention versus fluent wrong values—is a property of the eviction policy that averaged accuracy is structurally unable to report. The closest conceptual neighbors are selective-prediction analyses of QA (Geifman and El-Yaniv, 2017; Kamath et al., 2020),

which study the answer/abstain decision for uncompressed models; we bring that lens to the compression setting where, as we show, the operating point is not fixed but moved by the serving stack.

Query visibility. That compression behaves differently with and without sight of the query is established: Ada-KV evaluates question-aware and question-agnostic protocols on RULER (Feng et al., 2024); SCBench documents query-dependent long-context methods degrading when the query is unavailable (Li et al., 2025); and, concurrently with this work, Luo et al. (2026) audit six methods at matched budgets under both protocols, finding rank reversals that include SnapKV. Section 4.3 is accordingly a replication and extension of this line in the multilingual ONERULER setting; what it adds is the failure-composition dimension—precision-given-answer and value origin—that ranking audits do not report.

Multilingual long context. ONERULER (Kim et al., 2025) extends RULER (Hsieh et al., 2024) to 26 languages and introduces the absent-needle variant; its authors already observe that models over-predict absence at baseline. KV-CoRE (J. Chen et al., 2026) benchmarks the data-dependent low-rank compressibility of KV caches across sixteen languages, finding language-linked structure at the representation level; it does not evaluate eviction policies on task accuracy. To our knowledge, prior work has not jointly reported present/absent operating points, precision-given-answer, and per-language generator/scorer audits for KV-cache eviction, and we are not aware of a prior audit of the ONERULER generator’s per-language output or scorer.

3 Experimental Setup

Models. Llama-3.1-8B-Instruct (Grattafiori et al., 2024) and Qwen2.5-7B-Instruct-1M (Qwen Team, 2025), greedy decoding, 64 new tokens (raised from the generator’s 30-token metadata default to accommodate verbose languages), each model’s own chat template applied identically across all conditions.

Data. We generate *niah_single* (queried needle present, no other needles) and *niah_none* (four distractor needles present, queried key absent) with the official ONERULER generator at its released defaults for these tasks (book haystacks; 7-digit numeric values; seed 42), in seven languages spanning scripts and resource levels: en, pl, zh, ja, ko, vi, sw. The seven are the survivors of a baseline floor filter applied before the compression sweep: Hindi (0.00), Russian (0.30), and Tamil (0.53) were excluded for weak uncompressed performance, so language selection was conditioned on baseline results; the same set is used for both models, and there was no preregistration. Contexts are length-matched to 32K tokens under the Llama tokenizer and shared verbatim by both models; 100 items per language $\times$ task (1,400 unique contexts). Needle depth is sampled per item. Instructions in *both* tasks tell the model to output the language’s none-word if no value exists, so

abstention is a licensed response, not a refusal.

Detection-task composition. The two tasks differ in more than target presence: a present-trial context contains the queried needle and nothing else needle-like, while an absent-trial context contains four distractors. *H* and *FA* are therefore answer rates on two related but compositionally different tasks, and throughout the paper *d'* and *c* are descriptive indices under the equal-variance Gaussian model, not identified sensitivity and criterion. The mismatch has at least one mechanical consequence, flagged where it matters (§4): under positional eviction an absent context offers four needles that can survive the retained window against the present context’s one. A matched design—identical distractors, target inserted or omitted—would remove the confound and is the natural next step.

Compression. Two prefill-compression protocols. The primary protocol is *query-agnostic*, matching the public kvpress leaderboard: the policy scores and evicts only context tokens; the question is prefilled afterwards against the compressed cache and is never compressed. This models reusable document caches compressed before any question exists. The second is a *query-aware variant*: the whole templated prompt, question included, is prefilled under the policy, so SnapKV’s trailing observation window sits on the question. It is close to but not identical to canonical query-aware operation—KNorm may evict question tokens there, and the budget denominator is the prompt rather than the context, a difference of $\sim 0.4\%$ at these lengths. Policies: KNorm, SnapKV, StreamingLLM at kvpress 0.5.4 defaults (SnapKV: observation window 64, kernel 5; StreamingLLM: 4 sink tokens plus a recency window; KNorm: key- ℓ_2 scoring); budgets keep 25% or 50% of KV entries, plus the uncompressed baseline. The query-agnostic protocol covers both tasks (2 models $\times$ 7 conditions $\times$ 2 tasks $\times$ 7 languages $\times$ 100 items = 19,600 generations); the query-aware variant covers every *niah_single* condition (a further 9,800), serving both the protocol contrast of §4.3 and the determinism control of §6, for 29,400 total.

Response coding. We deliberately do not use the benchmark scorer to define the detection response (Section 5 shows why). A trial is coded *answered* iff the NFKC-normalized output contains a digit run of length ≥ 6 (needle values are 7-digit; the floor excludes years and enumeration artifacts). Correctness of answered present-trials uses the official exact-match scorer. We verified that every officially-correct present trial is coded answered, so the coding is a strict superset of correct responses.

Statistics. *H* and *FA* are macro-averaged over languages; $z(\cdot)$ is clipped to $[1/2n, 1 - 1/2n]$ with $n=100$. Precision-given-answer pools items across languages. Uncertainty is a $B=3000$ item bootstrap, resampled within language and *paired* across conditions (baseline and compressed cells share items), reported as 95% percentile intervals; value-origin proportions carry Wilson score intervals (bracketed percentages). All tables and figures are regenerated from

per-item records by a released script.

4 A Signal-Detection View of Compressed Retrieval

4.1 The criterion index moves conservative in every tested condition; the sensitivity index falls in nearly all

Table 1 gives the full operating points; Figure 1 plots them. Three regularities.

First, *every* compressed condition moves the criterion index conservative: Δc ranges from +0.17 (Llama KNorm-50%) to +1.89 (Qwen KNorm-25%), and all twelve paired CIs exclude zero. Compressed models answer less on present trials and, mostly, also less on absent trials. The direction is intuitive in hindsight—evidence for the needle is degraded, and both models were trained on instructions that make “none” available—but we are not aware of it having been measured in this operating-point form (the nearest observation we know of is by [Ananthanarayanan et al. \(2026\)](#), who find *query-conditioned* pruning suppresses correct abstention on an unanswerable coreference probe: a complementary regime and the opposite direction), and its consistency across a position-based, an attention-based, and an embedding-based policy suggests it is not an artifact of any one scoring rule. Whether it extends beyond the tested models, policies, budgets, and protocol is untested.

Second, the shift is not the whole story: d' falls in 11/12 conditions, in the worst cases severely. Llama under StreamingLLM lands at $d' \approx -0.2$ at both budgets, and the inversion is reliable—the bootstrap CI on d' itself is $[-0.35, -0.09]$ at 25% and $[-0.33, -0.08]$ at 50%: the model answers *more* often on absent contexts than on present ones. Part of this is mechanical rather than cognitive: an absent context contains four needles, any of which can fall inside the retained window and trigger an answer, while a present context contains one (§3). The one exception is instructive—Llama KNorm-50% preserves detection exactly ($\Delta d' = +0.02$, CI $[-0.13, 0.17]$) while identification precision drops from 0.98 to 0.66. At moderate budgets, key-norm eviction damages the value without damaging the model’s sense that a value is there.

Third, the two models start from different criteria and compression respects that: Qwen begins near-neutral ($c = -0.30$, FA = 0.22) and is pushed deep into conservatism (c up to +1.6); Llama begins liberal ($c = -0.83$, volunteering values on 57% of absent trials, consistent with over-answering being one of the baseline pathologies the ONERULER authors describe from the opposite direction ([Kim et al., 2025](#))) and is pushed toward neutral. Compression does not create the abstention behavior; it re-weights a disposition the instruction-tuned model already has.

4.2 The dissociation: similar accuracy, opposite failure

The decomposition’s payoff is Figure 2. On Llama at a 25% budget, KNorm and StreamingLLM are 4 points apart in the leaderboard statistic (0.21 vs. 0.25). Underneath, KNorm answers on 62% of present trials and is wrong in two-thirds of those answers (precision 0.34 [0.30, 0.37]); StreamingLLM answers on 25% and is wrong in one-hundredth of them (0.99 [0.97, 1.00]). Qwen replicates the ordering at every budget: precision-when-answering is KNorm < SnapKV < StreamingLLM in all six within-model comparisons, with StreamingLLM at 0.98–0.99 everywhere under this protocol. A practitioner choosing between these policies by accuracy delta would see a tie; a practitioner whose downstream system silently consumes wrong values would not.

What the wrong values are: corrupted copies, not inventions

The absent trials sharpen the picture—though not in the direction an exact-match analysis alone would suggest. Each `niah_none` context contains four distractor key-value pairs; we classify every answered absent trial by the origin of its emitted values (Table 2; coding details and a chance-rate control in App. D). At baseline, 396/397 (Llama) and 155/155 (Qwen) of answered absent trials copy a distractor *exactly*, and exact copying stays dominant under StreamingLLM and SnapKV. Under KNorm at 25% the exact-copy rate collapses to 48% (167/348, Llama) and 5/13 (Qwen)—but the freed mass overwhelmingly stays anchored to the input: 176/348 (51%) of Llama’s answers are *near-copies* of a distractor. Two aggregations quantify what remains (both in Table 2). Under the mutually exclusive trial label, only 5/348 (1.4% [0.6, 3.3]) of trials emit no grounded value at all; existentially, 47/348 (13.5% [10.3, 17.5]) contain *at least one* value that appears nowhere in the input, typically a single ungrounded entry inside a longer list of copied and corrupted distractors—at the level of individual emitted values, 89% (679/763) remain exact or near-copies. The same signature dominates present trials (App. D, Table 8): of the 282 answered Llama KNorm-25% present trials whose output lacks the true value, 272 are near-copies of it, and 263 of those are truncations (tie-break convention in App. D)—the model emits the first six digits of the 7-digit needle and stops (e.g., gold 2867825 → output 286782); at baseline the truncation count is zero. Key-norm eviction at aggressive budgets does not untether the output from the context; it corrupts values that remain anchored to it. For a downstream consumer these silent near-misses are as wrong as inventions and harder to spot. The models differ in degree. On present trials, trials emitting only ungrounded values stay below 5% in every Llama condition but reach 26% [18, 36] under Qwen KNorm-25% and 28% [23, 33] under Qwen SnapKV-25%—Qwen does fabricate at a meaningful rate where Llama essentially does not. On absent trials the ex-

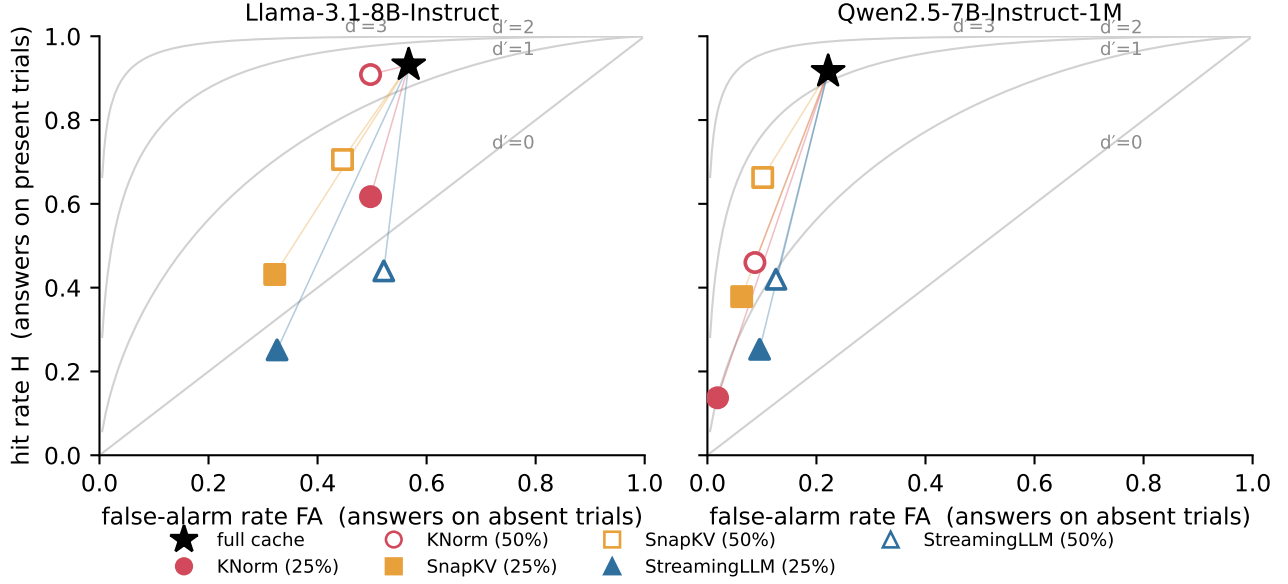


Figure 1: Compression moves the detection operating point; it does not slide along an iso-sensitivity curve. Each point is one policy×budget condition in (FA, H) space; gray curves are iso- d' loci, the diagonal is chance. Movement along a curve would be a pure criterion shift; movement across curves is sensitivity loss. All compressed conditions move down-left of baseline (more conservative), and all but Llama/KNorm-50% also cross to lower d' . Llama StreamingLLM lands below the diagonal: it answers more often on absent contexts than present ones (reliably so—the bootstrap CI on d' excludes 0; see §4), an inversion that is partly mechanical, since absent contexts contain four needles to the present context’s one (§3). Answer precision, as Table 1 shows, stays intact. Because the present and absent tasks differ in distractor composition (§3), the iso- d' curves are descriptive reference contours, not estimates from a matched signal/noise experiment.

istential rate is the sharper summary: at the 25% budget it reaches 13.5% for Llama under KNorm and 8.0% under SnapKV, and 9/13 (KNorm) and 15/44 (SnapKV) answered trials for Qwen. (The Qwen KNorm-25% absent-trial cell rests on $n=13$ answered trials; its intervals are correspondingly wide.)

4.3 Policy rankings and failure composition depend on query visibility

Everything above uses the query-agnostic protocol. The query-aware variant was generated for every `niah_single` condition, so present-trial accuracy and precision can be compared across protocols directly (Table 3). Three facts. First, SnapKV’s position in the ordering is a property of the protocol: given sight of the question, SnapKV is near-lossless at both budgets on both models (accuracy 0.90–0.92; +0.56 [0.53, 0.59] on Llama and +0.75 [0.72, 0.77] on Qwen at 25%). Its “intermediate” character elsewhere in this paper is a statement about query-agnostic serving—reusable caches compressed before the question exists—not about SnapKV as published; the protocol dependence itself has prior and concurrent support (Feng et al., 2024; Li et al., 2025; Luo et al., 2026). Second, the precision contrast survives query visibility: at 25%, KNorm precision is 0.23/0.06 (Llama/Qwen) against StreamingLLM’s 0.90/0.62, so corrupted-values-versus-silence is not an artifact of hiding the query. The matched-accuracy pairing

itself does not recur—query-aware KNorm falls to accuracy 0.12/0.01 while StreamingLLM holds at 0.23/0.25—so what is protocol-stable is the failure-mode ordering, not the accuracy tie. Third, precision itself is protocol-sensitive for the positional policy: Qwen StreamingLLM falls from 0.98 (query-agnostic) to 0.62 (query-aware)—“StreamingLLM answers with near-perfect precision” is a query-agnostic statement. KNorm accuracy is slightly *lower* query-aware on Llama (−0.08 to −0.11, CIs exclude 0), consistent with the variant’s ability to evict question tokens. Absent-trial behavior (FA) under the query-aware variant was not generated and is left to future work.

4.4 Mechanism, insofar as the controls license one

The pattern is consistent with the geometry of the policies. StreamingLLM’s retention set is a function of position only: a needle inside the retained window survives verbatim and is retrieved; a needle outside it is removed wholesale, leaving no corrupted trace to read out—hence inverted detection with near-perfect precision under the query-agnostic protocol. Our positional control supports this quantitatively: predicted accuracy = baseline accuracy × kept fraction matches observed StreamingLLM accuracy within 0.05 in 45/56 cells (mean |err| 0.036, max 0.13). KNorm, by contrast, retains a content-scored scatter that can preserve enough of the needle’s neighborhood to trigger an answer without preserving

Table 1: Detection and identification under eviction. Acc: official macro accuracy on present trials (the leaderboard number). H/FA : answer rates on present/absent trials under scorer-independent coding. Δc , $\Delta d'$: change vs. the same model’s baseline, paired item bootstrap [95% CI]. Prec: $P(\text{correct} \mid \text{answered, present})$, pooled [95% CI]. All twelve Δc are positive with CIs excluding 0; eleven of twelve $\Delta d'$ are negative with CIs excluding 0 (exception: Llama KNorm-50%).

Model	Condition	Acc	H	FA	d'	c	Δc [CI]	$\Delta d'$ [CI]	Prec [CI]
Llama-3.1-8B	full cache	0.91	0.93	0.57	1.32	-0.83	—	—	0.98 [0.97, 0.99]
	KNorm (25%)	0.21	0.62	0.50	0.31	-0.15	+0.68 [0.60, 0.77]	-1.01 [-1.20, -0.83]	0.34 [0.30, 0.37]
	KNorm (50%)	0.60	0.91	0.50	1.34	-0.66	+0.17 [0.09, 0.24]	+0.02 [-0.13, 0.17]	0.66 [0.63, 0.70]
	SnapKV (25%)	0.35	0.43	0.32	0.29	+0.32	+1.15 [1.06, 1.23]	-1.03 [-1.21, -0.86]	0.81 [0.77, 0.85]
	SnapKV (50%)	0.63	0.71	0.45	0.68	-0.20	+0.63 [0.55, 0.71]	-0.64 [-0.80, -0.49]	0.89 [0.87, 0.92]
	StreamingLLM (25%)	0.25	0.25	0.33	-0.22	+0.56	+1.39 [1.29, 1.50]	-1.54 [-1.76, -1.33]	0.99 [0.97, 1.00]
	StreamingLLM (50%)	0.43	0.44	0.52	-0.20	+0.05	+0.88 [0.78, 0.97]	-1.52 [-1.72, -1.34]	0.99 [0.97, 1.00]
Qwen2.5-7B-1M	full cache	0.91	0.92	0.22	2.14	-0.30	—	—	0.99 [0.98, 1.00]
	KNorm (25%)	0.02	0.14	0.02	0.99	+1.59	+1.89 [1.76, 2.05]	-1.15 [-1.41, -0.86]	0.12 [0.06, 0.19]
	KNorm (50%)	0.16	0.46	0.09	1.26	+0.73	+1.03 [0.95, 1.12]	-0.89 [-1.06, -0.72]	0.34 [0.30, 0.39]
	SnapKV (25%)	0.16	0.38	0.06	1.22	+0.92	+1.22 [1.14, 1.32]	-0.92 [-1.11, -0.72]	0.41 [0.36, 0.46]
	SnapKV (50%)	0.37	0.66	0.10	1.69	+0.43	+0.73 [0.65, 0.81]	-0.45 [-0.62, -0.28]	0.55 [0.51, 0.59]
	StreamingLLM (25%)	0.25	0.25	0.10	0.64	+0.99	+1.29 [1.19, 1.39]	-1.50 [-1.70, -1.31]	0.98 [0.95, 0.99]
	StreamingLLM (50%)	0.42	0.42	0.13	0.94	+0.67	+0.98 [0.90, 1.07]	-1.20 [-1.38, -1.02]	0.99 [0.98, 1.00]

Table 2: Value origin on answered absent trials (query-agnostic protocol). Reference set: every ≥ 6 -digit run of the item’s own serialized input—in practice the needle values (App. D). Exact/Near/Novel is the mutually exclusive *most-grounded* trial label: *exact* copy; *near*-copy (within Levenshtein distance 2 of an input value, or a ≥ 6 -digit truncation of one); *novel* (every emitted value matches nothing in the input). Any-nov.: trials containing *at least one* ungrounded value—the metric matching the phrase “appears nowhere in the input.” Both percentages carry Wilson 95% intervals. A random same-width value trips the near-copy rule 0.07% of the time (App. D), so the near tier is not an artifact of a loose rule.

Condition	n ans.	Exact	Near	Novel	Novel % [CI]	Any-nov. % [CI]
<i>Llama-3.1-8B</i>						
full cache	397	396	0	1	0.3 [0.0, 1.4]	0.8 [0.3, 2.2]
KNorm (25%)	348	167	176	5	1.4 [0.6, 3.3]	13.5 [10.3, 17.5]
KNorm (50%)	348	275	72	1	0.3 [0.1, 1.6]	11.2 [8.3, 15.0]
SnapKV (25%)	225	208	8	9	4.0 [2.1, 7.4]	8.0 [5.1, 12.3]
SnapKV (50%)	312	294	9	9	2.9 [1.5, 5.4]	11.2 [8.2, 15.2]
Streaming (25%)	228	227	0	1	0.4 [0.1, 2.4]	0.4 [0.1, 2.4]
Streaming (50%)	365	355	0	10	2.7 [1.5, 5.0]	3.3 [1.9, 5.7]
<i>Qwen2.5-7B-1M</i>						
full cache	155	155	0	0	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]
KNorm (25%)	13	5	6	2	15.4 [4.3, 42.2]	69.2 [42.4, 87.3]
KNorm (50%)	61	52	7	2	3.3 [0.9, 11.2]	9.8 [4.6, 19.8]
SnapKV (25%)	44	37	3	4	9.1 [3.6, 21.2]	34.1 [21.9, 48.9]
SnapKV (50%)	71	60	6	5	7.0 [3.0, 15.4]	14.1 [7.8, 24.0]
Streaming (25%)	67	67	0	0	0.0 [0.0, 5.4]	4.5 [1.5, 12.4]
Streaming (50%)	88	88	0	0	0.0 [0.0, 4.2]	2.3 [0.6, 7.9]

the value intact—the profile A. Chen et al. (2026) independently describe as K-Norm evicting with little positional bias but poor semantic selection. The value-origin analysis makes this concrete: what survives is usually enough to read out *most* of the value but not all of it, which is the truncation signature of §4.2. SnapKV, whose query-agnostic observation window sits at the end of the context, is intermediate on both models under this protocol and near-lossless when the window instead covers the question (§4.3). We present

Table 3: Query-agnostic (qa) vs. query-aware (aw) protocols, present trials. Present-task accuracy under each protocol, the paired item-bootstrap difference, and precision-given-answer. The query-aware variant compresses the whole prompt, placing SnapKV’s observation window on the question (§3); it is close to, not identical to, canonical query-aware operation.

Condition	Acc _{qa}	Acc _{aw}	Δ Acc [CI]	Prec _{qa}	Prec _{aw}
<i>Llama-3.1-8B</i>					
full cache	0.91	0.92	+0.00 [-0.01, +0.01]	0.98	0.98
KNorm (25%)	0.21	0.12	-0.08 [-0.11, -0.06]	0.34	0.23
KNorm (50%)	0.60	0.49	-0.11 [-0.14, -0.09]	0.66	0.62
SnapKV (25%)	0.35	0.91	+0.56 [+0.53, +0.59]	0.81	0.98
SnapKV (50%)	0.63	0.91	+0.28 [+0.25, +0.31]	0.89	0.98
Streaming (25%)	0.25	0.23	-0.02 [-0.03, -0.00]	0.99	0.90
Streaming (50%)	0.43	0.41	-0.03 [-0.04, -0.01]	0.99	0.93
<i>Qwen2.5-7B-1M</i>					
full cache	0.91	0.91	+0.00 [-0.01, +0.02]	0.99	0.98
KNorm (25%)	0.02	0.01	-0.00 [-0.02, +0.01]	0.12	0.06
KNorm (50%)	0.16	0.16	+0.01 [-0.01, +0.03]	0.34	0.30
SnapKV (25%)	0.16	0.90	+0.75 [+0.72, +0.77]	0.41	0.98
SnapKV (50%)	0.37	0.92	+0.56 [+0.52, +0.59]	0.55	0.99
Streaming (25%)	0.25	0.25	+0.01 [-0.00, +0.02]	0.98	0.62
Streaming (50%)	0.42	0.43	+0.02 [+0.01, +0.03]	0.99	0.83

this as an interpretation consistent with the controls, not a demonstrated mechanism.

5 Auditing the Benchmark

Averaged accuracy hides failure modes; it also hides benchmark defects. Two surfaced in per-language analysis, and both are quantitatively large. Both audits are post hoc—they were run after per-language results had been inspected—and should be read as exploratory.

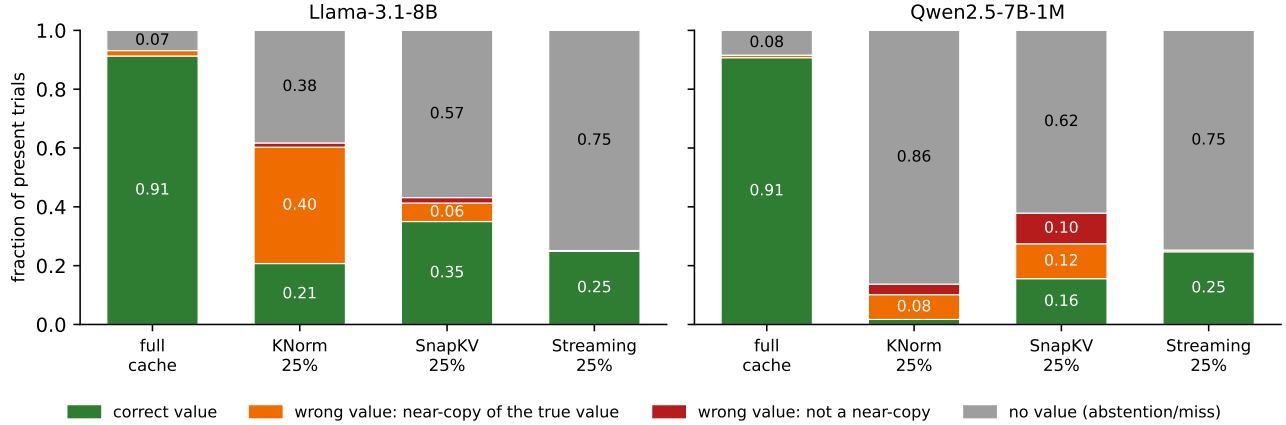


Figure 2: Outcome composition on present trials at a 25% budget. Present-task accuracy (green) is nearly matched for KNorm and StreamingLLM on Llama (0.21 vs. 0.25); the remaining mass is distributed in opposite ways—wrong values under KNorm, silence under StreamingLLM. Wrong values are split by origin (§4.2): near-copies of the true value (orange; under Llama KNorm-25% these are almost all truncations of its digits) vs. values that are not near-copies of it (dark red). The green band uses official scoring; a small number of answered trials contain the true value verbatim but fail the official exact-format scorer (5 under KNorm-25% and 23 under SnapKV-25% on Llama, from formatting or the none-word veto) and are counted in the orange band. Fractions ≥ 0.05 are labeled.

Table 4: Haystack redundancy by language (generator output at 32K, seed 42; means over 24 sampled contexts). Sent./Uniq.: sentences and unique sentences per context; Dup: duplicated-sentence fraction; Max: highest repeat count of a single sentence; Rep: implied mean repeats; D48: distinct character 48-gram ratio; gzip: compression ratio. All measures are tokenizer-free.

Lang	Sent.	Uniq.	Dup	Max	Rep	D48	gzip
en	1611	1546	0.040	44	1.04	1.000	0.384
pl	1186	1178	0.007	3	1.01	1.000	0.401
zh	1464	1460	0.003	2	1.00	1.000	0.445
ja	1264	1264	0.000	1	1.00	1.000	0.329
ko	1589	341	0.785	171	4.66	0.247	0.100
vi	1200	1027	0.144	93	1.17	1.000	0.331
sw	882	879	0.003	2	1.00	1.000	0.369

5.1 The Korean 32K haystacks are degenerate

Table 4: six of seven languages produce haystacks with essentially no repeated content (distinct 48-gram ratio ≥ 0.9998). Korean produces 1,589 sentences of which 341 are unique; the most frequent sentence appears 171 times; the text gzips to 10% of its size against 33–45% for every other language. Every model in this experiment—and any experiment using the generator at these settings—reads Korean “long context” that is one-quarter novel text.

The consequence is not neutral, because eviction policies differ in whether they can see it. Repeated sentences yield near-duplicate keys that content-based scorers can deprioritize almost for free, effectively concentrating the budget on the unique material that includes the needle; a position-based policy cannot. The data behave consistently with this reading (Table 5, right columns): Korean is Llama’s *worst* language uncompressed (0.77, rank 7/7) and becomes its

best or second-best under KNorm and SnapKV at both budgets, while dropping back to rank 6–7 under StreamingLLM. On Qwen, Korean holds rank 1 under every content-based condition and falls to rank 3–5 under StreamingLLM (ranks average over ties). We flag the causal status plainly: this is observational evidence, consistent with content-based policies benefiting from redundancy; we have not intervened on redundancy (deduplicated Korean, or repetition injected into another language) to demonstrate exploitation, and until such an intervention is run, “exploits” remains a hypothesis. What the observation establishes regardless is the audit point: a cross-language comparison that includes this split is partly measuring corpus degeneracy, and—because both models read the same generated text—cross-model replication cannot detect the problem.

Table 5 quantifies the stakes with the analysis a compression paper would naturally run: does compression amplify or suppress cross-language disparity? With all seven languages, Llama KNorm-25% amplifies overdispersion $5.2\times$ (paired CI excludes 1); excluding Korean, the estimate drops to 0.64 and is no longer distinguishable from no change. Qwen SnapKV-25% moves from 1.17—itself indistinguishable from 1—to 0.06, reliably below 1. Only two patterns survive the exclusion: SnapKV amplification on Llama (5.3 $\rightarrow$ 8.3, both CIs excluding 1) and StreamingLLM suppression on both models (0.06–0.21 throughout)—and the latter is largely mechanical, since the positional law of Section 4.4 rescales every language’s accuracy by the same kept fraction, and ϕ is not invariant to such rescaling. We therefore make no headline claim about disparity amplification ourselves; the point of Table 5 is that without the per-language audit we would have.

Table 5: A single defective split can invert dispersion conclusions. Amp: ratio of cross-language accuracy overdispersion (ϕ , the $\chi^2/(k-1)$ inflation factor over a pooled-rate binomial) compressed vs. baseline, computed on all seven languages and with Korean excluded. **Bold:** the paired 95% bootstrap CI on the ratio excludes 1; non-bold values are indistinguishable from no change in dispersion. Right: Korean’s accuracy and rank (1 = best of 7, averaged over ties) in each condition; its baseline ranks are 7 (Llama, 0.77) and 1 (Qwen, 1.00).

Condition	Amp _{all}	Amp _{-ko}	ko acc	ko rank
<i>Llama-3.1-8B</i>				
KNorm (25%)	5.20	0.64	0.82	1
KNorm (50%)	2.26	2.63	0.87	1
SnapKV (25%)	5.28	8.32	0.62	2
SnapKV (50%)	4.34	6.82	0.78	3
Streaming (25%)	0.07	0.09	0.21	6
Streaming (50%)	0.06	0.06	0.38	7
<i>Qwen2.5-7B-1M</i>				
KNorm (25%)	0.03	0.03	0.04	1
KNorm (50%)	0.86	0.14	0.64	1
SnapKV (25%)	1.17	0.06	0.73	1
SnapKV (50%)	0.83	0.25	0.94	1
Streaming (25%)	0.08	0.09	0.27	5
Streaming (50%)	0.18	0.21	0.48	3

5.2 The absent-needle scorer applies different criteria to different models

The official `niah_none` scorer marks a response correct if the language’s none-word appears anywhere in it as a (whitespace-stripped) substring. Nothing penalizes values emitted alongside it. Llama’s modal Korean output at baseline is, verbatim in structure,

```
<answer> 4744854, 5108603, 3341057,
5614226, <none-word> </answer>
```

—the four distractor values of that item, then the none-word—and it scores as a correct rejection. Table 6: 85.5% of Llama’s officially-credited Korean responses and 71.4% of its Swahili ones contain a ≥ 6 -digit value; official 0.83 becomes strict 0.12 for Korean, 0.21 becomes 0.06 for Swahili, with a moderate Japanese effect (0.51 $\rightarrow$ 0.44) plausibly aggravated by the none-word being substring-prone. Qwen, which emits clean abstentions, moves by at most 0.09 (Swahili). Of the 98 query-agnostic absent-task cells—the only cells this defect can touch—12 shift by ≥ 0.05 under strict scoring and 7 by ≥ 0.10 ; 11 of the 12 involve Llama.¹

The defect is therefore not a uniform bias but a model-dependent one: the benchmark’s absent-needle number measures genuine abstention for one model and value-plus-none-word co-occurrence for another. Any cross-model comparison on this task inherits the gap, and any compression

¹A separate *lenient* variant of the present-task scorer (number match without the none-word veto) moves 11 of the 196 present-task cells by ≥ 0.05 . It probes formatting strictness rather than false presence and is reported separately in the released analysis; v1 of this paper summed the two populations into a single count of 23/294.

Table 6: Official vs. strict scoring of `niah_none`, uncompressed baseline. Strict = official *and* no ≥ 6 -digit value anywhere in the output. Contam.: share of officially-credited responses that contain such a value.

Lang	Official	Strict	Δ	Contam.
<i>Llama-3.1-8B</i>				
en	0.63	0.62	+0.01	1.6%
pl	0.93	0.91	+0.02	2.2%
zh	0.24	0.22	+0.02	8.3%
ja	0.51	0.44	+0.07	13.7%
ko	0.83	0.12	+0.71	85.5%
vi	0.63	0.62	+0.01	1.6%
sw	0.21	0.06	+0.15	71.4%
<i>Qwen2.5-7B-1M</i>				
en	0.96	0.96	+0.00	0.0%
pl	1.00	1.00	+0.00	0.0%
zh	0.94	0.94	+0.00	0.0%
ja	0.75	0.71	+0.04	5.3%
ko	0.37	0.36	+0.01	2.7%
vi	0.90	0.90	+0.00	0.0%
sw	0.58	0.49	+0.09	15.5%

study using the official scorer will *understate* compression-induced false presence exactly where co-emitted values are most likely. Our own analysis is immune by construction—the detection coding of Section 3 ignores the scorer—which is also why we recommend it: on this task, “emits a candidate value” is the behaviorally meaningful event, and it is checkable by regular expression.

6 Validity Controls

Positional law. For StreamingLLM, retention is a deterministic function of depth, and needle depth is uniform by construction, so accuracy should approximate baseline $\times$ kept fraction. It does: 45/56 cells within 0.05, mean $|\text{err}|$ 0.036, max 0.13. This validates the budget accounting and underwrites the mechanistic reading in Section 4.4.

Response-coding robustness. The coding’s failure modes are measurable on the full corpus. The superset property of §3—every officially-correct present trial is coded *answered*—holds with zero violations across all 29,400 outputs. A separator-tolerant variant of the value regex (additionally accepting 1,234,567-style formatting) changes the answered label of 26 outputs (0.09%); the largest resulting change in any macro-averaged H or FA is 0.007 (Llama SnapKV-25% H : 0.43 $\rightarrow$ 0.44), and no other Table 1 entry moves at reported precision.

Determinism. `niah_single` at `press=none` was run under two protocol variants sharing identical token IDs, so outputs should be bit-identical. They are not quite: 29/1,400 items (2.1%) flip correctness, concentrated in Qwen/Swahili (14/100, directional 11–3), an inference-stack nondeterminism we could not eliminate. This is a single paired comparison, so we report it as *measured* run-to-run instability under

this stack rather than a universal noise floor; per-cell claims of this magnitude should be read accordingly. Swahili—the flakiest cell—is also Qwen’s sole weak baseline language (0.48), so Qwen’s baseline cross-language dispersion should be read with corresponding caution.

Scorer independence. All detection quantities use the regular-expression coding; official scoring enters only through identification correctness on present trials, where it is an exact-match test on 7-digit values and the none-word confound of Section 5 cannot fire (a credited response cannot contain the none-word there).

7 Discussion and Recommendations

For practitioners. At similar accuracy, eviction policies are not interchangeable: position-based eviction converts missing evidence into visible abstention; key-norm eviction converts it into fluent wrong values that are usually corrupted copies of true context values—silent near-misses that no downstream consumer can distinguish from correct retrievals; SnapKV sits between under query-agnostic serving and is near-lossless when the query is visible to the policy (§4.3). If downstream consumers cannot verify values, the precision column of Table 1 is the safety-relevant number—read jointly with coverage, since StreamingLLM’s 0.99 precision comes at answering on only 25% of present trials—and it varies by up to 65 points between policies that a leaderboard ranks as equivalent. Which trade-off is preferable depends on the relative cost of a wrong value versus an abstention; we make no universal safety claim.

For benchmark and leaderboard authors. (i) Report present/absent tasks as an operating point— H , FA, and precision-given-answer—not as two accuracies averaged into one. (ii) Score abstention strictly: a none-response containing a candidate value is a false presence, not a correct rejection. (iii) Audit generated corpora per language before cross-language claims; a tokenizer-free redundancy check (unique sentences, distinct n -grams, gzip ratio) takes minutes and, in our case, changes the sign of downstream conclusions.

What we do not claim. We do not claim that d' and c identify an internal sensitivity or decision threshold—the present and absent tasks differ in composition (§3), so they are descriptive indices of two answer rates. We do not claim the tested query-aware variant equals canonical query-aware serving (§3). We do not claim compression harms or helps cross-language fairness—Table 5 shows our design cannot cleanly answer that with a defective split and $k=7$ languages—and Appendix B shows the same about tokenizer fertility as an explanatory variable. We also do not claim the criterion movement is desirable even though conservatism sounds safe: on Qwen KNorm-25%, the shift co-occurs with precision of 0.13, the worst of both worlds.

8 Limitations

Two instruction-tuned 7–8B models; one context length (32K); one synthetic task family; eviction only (no quantization or low-rank methods). The present and absent tasks differ in composition, so the SDT summaries are descriptive indices (§3); a matched present/absent design is the natural fix. The seven languages are the survivors of a baseline floor filter (§3), so language selection was conditioned on baseline results, and the benchmark audits of §5 are post hoc; none of this was preregistered. The redundancy–eviction link is observational; no intervention on redundancy was run. A single generator seed, so the Korean degeneracy is established for the generator at these settings and seed rather than for the released ONERULER data, and should be checked against the published split, additional seeds, and other context lengths before being attributed to the benchmark at large. The query-aware protocol variant is close to but not identical to canonical query-aware serving (§3), and its absent-trial behavior was not generated. Response coding is heuristic, though its measured failure modes are small (§6). All conditions were run once per protocol (single seed); the determinism control bounds only run-to-run instability of the baseline present task. SDT summaries use the equal-variance Gaussian model with clipping at extreme rates; macro-averaging over seven languages weights them equally regardless of baseline difficulty. Precision is pooled across languages and can mask per-language heterogeneity (Appendix A).

Reproducibility

Data were produced with the official ONERULER generator (niah_single: one needle; niah_none: four distractor needles, queried key absent; book haystacks, 7-digit values, seed 42), length-matched to 32K with the Llama tokenizer and shared across models: 1,400 unique contexts, 29,400 generations. Inference: greedy decoding, 64 new tokens, bfloat16, SDPA attention, single NVIDIA L40S; torch 2.13.0, transformers 5.2.0, kvpress 0.5.4, datasets 5.0.0; HF model revisions 0e9e39f2 (Llama-3.1-8B-Instruct) and e28526f7 (Qwen2.5-7B-Instruct-1M); press hyperparameters at kvpress defaults (§3). Detection coding: NFKC-normalize, then `\d{6,}`. Bootstrap: $B=3000$, resampled within language, indices shared across paired conditions. Per-item records for all 29,400 generations, the sweep code (`modal_kv_audit_v4.py`), and the analysis scripts that regenerate every derived table and figure in this paper (`post_run_calc.py`, `make_tables.py`) are available at <https://github.com/elad-simbalista/kv-cache-eviction>. The kvpress version is 0.5.4; its exact commit hash was not recorded.

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A Per-Language Operating Points

Table 7 reports per-language H/FA at the aggressive budget. Two cells deserve note. Llama/Swahili is the one language where the baseline itself barely performs the absent-needle task (strict correct rejection 0.06; Table 6), so its detection numbers are dominated by the scorer pathology rather than by compression. Qwen/Swahili is that model’s sole weak present-trial language at baseline ($H=0.50$ vs. ≥ 0.96 elsewhere) and also the least deterministic cell (Section 6); Qwen’s large baseline cross-language dispersion is essentially a Swahili effect.

Table 7: H/FA per language, baseline and 25% budget (scorer-independent coding).

Model	Condition	en	pl	zh	ja	ko	vi	sw
Llama-3.1-8B	full cache	1.00/0.38	0.84/0.09	0.98/0.76	0.98/0.56	0.85/0.87	0.90/0.38	0.97/0.93
	KNorm (25%)	0.75/0.28	0.58/0.10	0.36/0.44	0.70/0.52	0.85/0.87	0.72/0.66	0.36/0.61
	SnapKV (25%)	0.52/0.05	0.04/0.02	0.25/0.24	0.43/0.11	0.79/0.85	0.05/0.03	0.94/0.95
	StreamingLLM (25%)	0.27/0.18	0.28/0.24	0.27/0.33	0.25/0.12	0.23/0.59	0.19/0.23	0.27/0.59
Qwen2.5-7B-1M	full cache	0.98/0.04	0.99/0.00	0.98/0.06	1.00/0.28	1.00/0.62	0.96/0.10	0.50/0.45
	KNorm (25%)	0.14/0.01	0.06/0.00	0.11/0.00	0.15/0.00	0.35/0.08	0.15/0.00	0.00/0.04
	SnapKV (25%)	0.62/0.03	0.20/0.00	0.34/0.01	0.43/0.02	0.84/0.31	0.12/0.02	0.10/0.05
	StreamingLLM (25%)	0.27/0.06	0.28/0.00	0.27/0.02	0.28/0.11	0.28/0.44	0.29/0.03	0.10/0.01

B Tokenizer Fertility Is Not a Usable Explanation Here

A natural hypothesis is that cross-language differences under compression track tokenizer fertility (tokens per byte), since higher-fertility languages spend more of a fixed token budget per unit content. We tested Spearman correlations between per-language fertility (measured on FLORES and on the evaluation corpora; the two agree, $\rho \approx 0.86$ between models’ fertility vectors) and per-language accuracy or retention across all 72 condition $\times$ outcome combinations. None survives Benjamini–Hochberg correction; signs split 40 negative / 30 positive / 2 zero. Most tellingly, the single strongest correlation in each model occurs in the *identical* condition (niah_single, StreamingLLM-25%, retention) with *opposite* signs: $\rho = +0.93$ on Llama and $\rho = -0.74$ on Qwen. We do not read this as evidence of no fertility effect: with $k=7$ languages the design’s power to detect even $\rho = 0.7$ is roughly 0.37 (0.53 at $\rho = 0.8$) by simulation, and under StreamingLLM the retention outcome is nearly constant across languages, making ranks unstable. The correct conclusion is that this design cannot resolve fertility effects, and the sign instability across models on the same predictor means single-model fertility correlations in this regime should not be interpreted at all.

C Additional Reproducibility Details

Cells are fully crossed: for each model, 7 conditions (baseline; {KNorm, SnapKV, StreamingLLM} $\times$ {25%, 50%}) $\times$ 2 tasks $\times$ 7 languages $\times$ 100 items, with the same 100 items reused across conditions within each language $\times$ task (enabling the paired bootstrap). Budgets are fractions of KV entries; measured kept fractions were asserted against tolerance at run time. The overdispersion statistic of Table 5 is $\phi = \frac{1}{k-1} \sum_{\ell} \frac{(x_{\ell} - n\bar{p})^2}{n\bar{p}(1-\bar{p})}$ over $k=7$ languages with $n=100$; note ϕ is not invariant to uniform rescaling of accuracies, which is one reason the StreamingLLM suppression values in Table 5 should be read qualitatively. *Exact-copy* in §4.2 means a canonicalized ≥ 6 -digit value in the output equals a ≥ 6 -digit digit run of that item’s full serialized input; on these tasks such runs are, in practice, the needle values themselves (the four distractors on absent trials, plus the target on present trials), since the book haystacks contribute essentially no ≥ 6 -digit runs (empirically: across all 19,576 candidate values emitted in the corpus, none matched an input digit run other than the needle values). Baseline exact-copy rates on answered absent trials are 99.7% (396/397, Llama) and 100% (155/155, Qwen); the KNorm-25% rates discussed in the text are 48.0% (167/348, Llama) and 5/13 (Qwen).

D Value-Origin Coding

Every ≥ 6 -digit digit run in an output (after NFKC normalization) is canonicalized by stripping leading zeros and classified against a per-item reference set: every ≥ 6 -digit run of the item’s own serialized input—in practice its target value (present trials) and its four distractor values (App. C). Each trial then carries two aggregations. The mutually exclusive taxonomy of Tables 2 and 8 assigns the *most-grounded* label among the trial’s emitted values, in the order exact $>$ near-copy $>$ novel; a trial is Novel only if *every* value it emits is ungrounded. The *Any-nov.* columns report the existential aggregation—the trial emitted at least one ungrounded value—and only that metric supports the phrase “appears nowhere in the input.”

Near-copy means Levenshtein distance ≤ 2 from a reference value, or one string being a ≥ 6 -digit prefix or suffix of the other (truncation). A ≥ 6 -digit prefix of a 7-digit value is also within Levenshtein distance 2 of it; where both clauses fire we count the truncation reading, which yields the 263 of §4.2 (a distance-first convention counts 262). The rule is deliberately tight: drawing values uniformly at random with the needles’ digit-width, the probability that a random value trips the near-copy rule against an item’s reference set is 0.07% (14/20,000 Monte-Carlo draws; Wilson 95% [0.04, 0.12]), so the near-copy tier cannot be explained as a loose-matching artifact. As a further check, matching each output against the pooled numeral inventory of *other* items in the same language classifies at most 6 of 29,400 outputs differently, so per-item and corpus-level reference sets agree in practice. Table 8 reports the present-trial taxonomy; “exact” there means the output contains the true value (possibly alongside others, so the count can slightly exceed official accuracy, which additionally requires the none-word veto and exact answer formatting).

Table 8: Value origin on *answered present* trials (query-agnostic protocol), classified and aggregated as in Table 2. Under Llama KNorm-25%, 263 of the 272 near-copies are truncations of the target.

Condition	<i>n</i> ans.	Exact	Near	Novel	Novel % [CI]	Any-nov. % [CI]
<i>Llama-3.1-8B</i>						
full cache	652	652	0	0	0.0 [0.0,0.6]	0.0 [0.0,0.6]
KNorm (25%)	432	150	272	10	2.3 [1.3,4.2]	3.0 [1.8,5.1]
KNorm (50%)	636	424	209	3	0.5 [0.2,1.4]	0.6 [0.2,1.6]
SnapKV (25%)	302	268	21	13	4.3 [2.5,7.2]	4.3 [2.5,7.2]
SnapKV (50%)	494	459	26	9	1.8 [1.0,3.4]	1.8 [1.0,3.4]
Streaming (25%)	176	174	0	2	1.1 [0.3,4.0]	1.1 [0.3,4.0]
Streaming (50%)	308	306	0	2	0.6 [0.2,2.3]	0.6 [0.2,2.3]
<i>Qwen2.5-7B-1M</i>						
full cache	641	640	1	0	0.0 [0.0,0.6]	0.0 [0.0,0.6]
KNorm (25%)	96	13	58	25	26.0 [18.3,35.6]	27.1 [19.2,36.7]
KNorm (50%)	322	112	179	31	9.6 [6.9,13.3]	9.6 [6.9,13.3]
SnapKV (25%)	265	110	82	73	27.5 [22.5,33.2]	27.5 [22.5,33.2]
SnapKV (50%)	464	258	133	73	15.7 [12.7,19.3]	16.2 [13.1,19.8]
Streaming (25%)	177	177	0	0	0.0 [0.0,2.1]	0.0 [0.0,2.1]
Streaming (50%)	294	294	0	0	0.0 [0.0,1.3]	0.0 [0.0,1.3]